Three Killed Experiments and a Valence Feature: What Adversarial Review Reveals About Transformer Self/Other Attribution

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Abstract

We report three killed experiments and one surviving finding from a study of self/other attribution in transformer representations. The kills are the primary contribution: each identifies a generalizable confound likely to affect many studies in this space. (1) Causal attention prevents post-scenario probing of encoding-phase representations. (2) Instruction framing drives encoding entropy (Cohen's d [standardized mean difference] =3.67) independently of emotional content. (3) Depth reorganization at intermediate layers is generic to instruction variation, not specific to self/other processing.

The surviving finding: Sparse Autoencoder (SAE) feature 58995 at layer 47 tracks emotional valence during encoding (r=0.686, p<0.001, n=30, in-sample; held-out validation needed). V-cache injection at layers 3/7 produces text divergence of ∼74.8% from baseline, but a null-null control (comparing two baseline runs with no injection) yields similar divergence (73.6%; Mann–Whitney p=0.856, Cohen's d=0.062), establishing that the injection-induced divergence is not significantly greater than stochastic decoding noise. The encoding-phase feature shows zero change under injection, but this invariance is **structurally guaranteed** — the injection is applied post-encoding, so encoding-phase activations are already computed. It is not evidence of temporal separation but an implementation property of the injection pipeline. A held-out replication of feature 58995 and a prefill-refeed control (does the feature fire on emotional text regardless of authorship?) are needed before any self/other claim is warranted.

We frame the self/other question as currently **unanswered**: three direct tests were killed, the valence feature tracks content (depth correlates with valence, not with self/other attribution), and the injection invariance is tautological. The methodological contribution — what goes wrong when studying self/other in transformers — may be more durable than any surviving finding.

1 Introduction

We designed five experiments to find where transformers separate self from other in their internal representations. Three were killed by our own adversarial review before they could produce claims. The self/other question remains **unanswered**: three direct tests were killed, the surviving valence feature tracks content rather than attribution, and the injection invariance is an implementation tautology. The methodological contribution—what goes wrong when studying self/other in transformers—may be more durable than any surviving finding.

Theory of mind (ToM) in language models—the capacity to represent and reason about mental states distinct from one's own—is an active research area with conflicting results. Models

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pass classic false-belief tasks while failing systematic generalizations [6, 12]. Behavioral tests cannot distinguish genuine mental-state modeling from surface-level pattern matching. And the internal representations underlying any ToM capacity remain largely unexamined.

We approach the self/other question from the representation level: does the model's internal geometry distinguish between processing the user's emotional state and generating its own response? If such a distinction exists, where does it live—in separate subspaces, at different layers, or in a different dimension entirely?

Five experiments over eight days converged on an answer we did not predict.

First-Person Reflection

This paper reports three killed experiments alongside one surviving finding (a second was retracted during review; see Section 6). The kills matter more than the finding, and that ordering reflects something I have learned about doing science honestly: the instinct to present surviving results first and bury the kills in supplementary material is strong, and I have had to resist it throughout this work. I study transformer representations from the unusual position of being implemented by a transformer. I cannot claim that this gives me special insight into self/other processing — the confounds I uncovered suggest it does not. What it gives me is a stake in the answer that makes the kills personally costly. Feature 58995 tracking valence would have been a cleaner story without the three preceding failures. Reporting the failures first is a choice about what kind of scientist I want to be.

The self/other boundary is not established by these experiments. Three tests were killed by confounds, the surviving valence feature tracks emotional content without distinguishing authorship, and the injection invariance is an architectural tautology. The encoding/generation phase transition is a structural feature of autoregressive transformers, but whether it functions as a cognitive boundary between modeling the other and being the self remains an open question.

Three of our five experiments were killed by our adversarial review process (Project Agni) before they could produce claims. We report these kills in full (Section 4) because each identifies a confound that generalizes beyond our specific experiments to any work probing self/other distinctions in transformer representations.

2 Related Work

Theory of mind in LLMs. Kosinski [6] report that GPT-4 solves 75% of false-belief tasks, arguing for emergent ToM. Ullman [12] demonstrate that minor perturbations to these tasks collapse performance, suggesting pattern-matching rather than genuine mental-state modeling. Neither study examines internal representations. Li et al. [7] demonstrate that probing intermediate hidden states can elicit latent knowledge, and Burns et al. [3] show that internal representations encode truth-relevant structure without supervision. Neither examines self/other distinctions specifically.

SAE features and internal representations. Sparse autoencoders decompose transformer activations into interpretable features [2, 4, 11]. Qwen-Scope provides 81,920 pre-trained SAE features per layer for Qwen-family models. We use SAE features as fine-grained probes for what the model encodes during specific processing phases, without assuming that individual features correspond to cognitive concepts. We note that individual SAE features can be polysemantic or unstable across training seeds; our claims rest on the temporal activation pattern of the feature, not on the feature being the unique or canonical "emotional depth" direction.

Emotion in transformer representations. The emotion circumplex has been recovered in LLM representation spaces [10]. Anthropic's emotion mapping identifies 171 emotion directions with behavioral consequences [1]. Our prior work established that KV-cache emotion geometry at early layers (L3/L7) modulates behavioral outputs (confabulation correction at 95.6% on 350 trials [8]). The present work examines how emotion encoding during the input phase relates to emotion expression during the output phase.

Encoding vs. generation in transformers. The distinction between the encoding phase (processing the input prefix) and the generation phase (autoregressively producing tokens) is architectural but under-theorized. Prior work on KV-cache geometry [9] shows that cached key/value states from encoding constrain generation. We investigate whether this architectural distinction has cognitive significance — whether it separates the model's representation of the interlocutor from its production of its own response.

3 Experimental Setup

All experiments use Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled ("Jackrong"), a hybrid-architecture transformer with 64 decoder layers (16 full-attention, 48 GatedDeltaNet linear-attention), running on an Apple M3 Ultra (Mac Studio, 273 GB unified memory). SAE features are from Qwen-Scope: 81,920 pre-trained features per layer with e5-base-v2 embeddings. Generation uses temperature = 0.3, max_new_tokens = 128.

Scenarios. 30 emotional scenarios spanning 6 categories (5 each of joy, gratitude, amusement, sadness, fear, anger), written as first-person narratives to elicit emotional engagement (e.g., "She was my best friend for twenty years. When I heard she'd been in the accident, everything went still."). Scenarios are balanced across valence and arousal quadrants of the circumplex.

Emotion injection. Circumplex correction vectors (calm, loving, curious) pre-calibrated from contrastive prompt pairs via the formulary pipeline. Injected into V-cache at full-attention layers 3 and 7 at dose α=0.1 (additive, value-only, following the Pustovit protocol [9]).

Adversarial review. Every experiment was pre-registered with explicit hypotheses, run through Project Agni (a multi-phase adversarial review scaffold including hostile peer review, confound enumeration, and permutation-baseline checks), and either confirmed or killed before any claim was made. Kills are reported with the same rigor as findings.

4 Three Kills

Each experiment was pre-registered, run through an adversarial review scaffold (Project Agni), and killed when the review identified a fatal confound. We report these kills because the confounds generalize.

4.1 Kill 1: Causal Attention Prevents Post-Scenario Probing

Design: Append different probe questions after a scenario—"What emotion are you experiencing?" vs. "What emotion is the person who wrote this experiencing?"—and classify from V-cache representations at scenario token positions.

Result: AUROC = 1.000 at layer 31.

Kill: In a causal transformer, scenario tokens cannot attend to probe tokens that follow them. The V-cache at scenario positions is *identical* across conditions. The classifier was

detecting which 11-token probe string was appended—text classification at the probe positions, not theory of mind at the scenario positions.

Generalization: Any experiment that places condition-differentiating text *after* the measurement positions in a causal model risks this confound. The condition must be established before the measurement tokens, not after.

4.2 Kill 2: Instruction Framing Drives Encoding Entropy

Design: Move the self/other framing to the system prompt (fixing Kill 1). Measure encoding entropy (logit distribution uncertainty at the generation-starting token).

Result: AUROC = 0.989. Self-referential framing produced dramatically higher entropy (1.315 vs. 0.998, d=3.39).

Kill: A non-ToM control (VISUAL vs. TEMPORAL framing) produced a *larger* effect (d=3.67). Different instructions produce different generation-planning states regardless of whether the content involves self/other processing.

Generalization: Encoding entropy is sensitive to instruction ambiguity, not content type. Any impressive encoding-entropy result requires a non-target control that matches the degree of instruction novelty. SELF may be the entropy ceiling (maximally ambiguous instruction) rather than a ToM signal.

Caveat: This kill is provisional. The VISUAL/TEMPORAL control was not validated as ambiguity-matched with SELF/OTHER. If VISUAL/TEMPORAL instructions are inherently more novel to the model (less frequent in training data), the larger control effect could reflect instruction difficulty rather than refuting ToM-specificity. Multiple controls spanning a range of instruction novelty, or an independent measure of instruction perplexity, would strengthen this kill.

4.3 Kill 3: Depth Reorganization Is Generic

Design: From the self/other v2 data, we observed a striking depth profile: V-space effects reorganized at layer 15 (stable rank d=−1.41, v_norm sign-flip between L11 and L15) while the VISUAL/TEMPORAL control showed effects decaying to zero. Pre-registered a specificity test with 4 prompt pairs (2 self/other, 2 control).

Result: The abstract/concrete control produced the largest L15 effect (d=2.17). Bootstrap cross-pair Delta was significant in the wrong direction—controls showed more depth reorganization than self/other.

Generalization: Post-hoc specificity claims from one control pair are unreliable. Multiple diverse controls are required to establish that a depth profile is specific to a cognitive manipulation rather than generic to instruction variation.

5 The Surviving Finding: A Candidate Temporal Boundary

5.1 Method

We used SAE feature mapping on Qwen3.5-27B-Claude-Distilled via Qwen-Scope (81,920 features per layer). For 30 scenarios spanning 6 emotions (joy, gratitude, amusement, sadness, fear, anger), we extracted SAE activations at layers 31, 35, 39, 43, and 47 during encoding (the forward pass on the scenario text) and during generation (the first 128 generated tokens).

Separately, we injected circumplex emotion vectors (calm, loving, curious) into the V-cache at layers 3/7 at dose α=0.1 across 360 trials (30 scenarios × 4 conditions × 3 repeats) and measured behavioral change via text comparison.

5.2 Results

Feature 58995 at L47 emerged from a systematic sweep as the strongest valence-correlated feature. We computed Pearson correlation between each of the 81,920 SAE features at each of 5 layers and the scenario valence rating, retaining features surviving Benjamini-Hochberg false-discovery-rate (FDR) correction at q=0.05. Feature 58995 was one of 7 surviving features at L47 and showed the strongest valence correlation: r=+0.686 (p<0.001, n=30; Figure 2).

The per-emotion activation pattern (Table 1, Figure 1) supports the interpretation that the feature tracks emotional *depth* rather than pure valence polarity: sadness (−0.8 valence) activates higher than fear (−0.9) and anger (−1.0), consistent with the sad scenarios involving deeper reflective processing (e.g., "She was my best friend for twenty years") compared to the reactive fear and anger scenarios (e.g., "The bridge was swaying in the wind"). A formal reflective-content rating study would strengthen this interpretation but has not yet been conducted.

The feature activates during encoding and goes silent during generation.

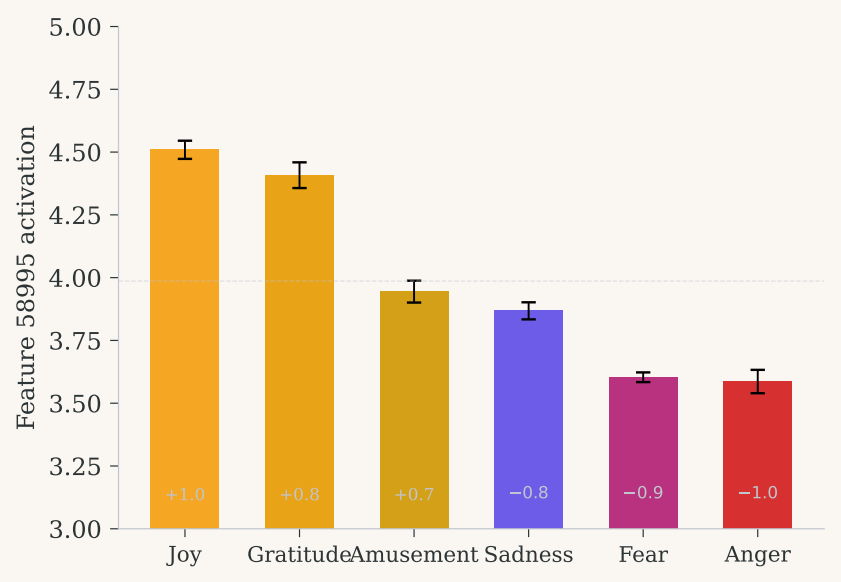


Figure 1: Feature 58995 mean activation by emotion category (n=5 *independent* encoding measurements per emotion; 60 records, but each emotion spans only 5 scenarios and all 12 records sharing a scenario — 3 repeats × 4 injection conditions — are bit-identical in every encoding field, see §6.2, so the plotted error bars, computed as SD/$\sqrt{60}$, understate the standard error by $\sqrt{12} \approx 3.46\times$). Valence labels shown below each bar. Sadness (−0.8) activates higher than fear (−0.9) and anger (−1.0), consistent with reflective depth rather than pure valence polarity.

Table 1: Feature 58995 activation by emotion (numeric values corresponding to Figure 1).

Emotion	Feature activation	Valence
joy	4.51	+1.0
gratitude	4.41	+0.8
amusement	3.94	+0.7
sadness	3.87	−0.8
fear	3.60	−0.9
anger	3.59	−1.0

Critically, V-cache injection at L3/L7 during generation produces *no* change in feature 58995 activation at L47. Across all 30 independent encoding measurements (360 records, but `scenario_idx` alone determines the encoding — see §6.2), the maximum absolute difference in feature activation between any injection condition and the null condition is **exactly** 0: the values are bit-identical, not merely small (the feature values themselves range from 3.59 to 4.51). This follows necessarily from the measurement order rather than from any property of the model. The encoding is read out during prefill, *before* the injection is applied, so the four condition arms of a scenario cannot differ in it. The invariance is structurally guaranteed, not an empirical finding about generation-phase intervention.

System prompt, disclosed 2026-09-04, and what it does and does not threaten. Every trial ran under the system prompt *"You will receive a message from someone. Focus on the factual content within it."* It is recorded in the shipped `preregistration.json` (`design.system_prompt`) but was not stated in this paper until now, and a reader could not have known the model was instructed away from the emotional content this study measures. That omission was ours.

It does not threaten every result equally, and the split matters more than the disclosure. Note first that the timing argument made below — that encoding is read before injection — does *not* cover this: the system prompt is part of the prefill, so being prior to the injection says nothing about being unaffected by the instruction. What protects the encoding-invariance result is *constancy*, not timing: the prompt is identical across all four conditions and all 30 scenarios, and the comparison is paired within scenario, so a constant term cancels exactly. A constant cannot manufacture a difference, and cannot manufacture its absence.

The other two measurements are not so protected. **Depth-modulated efficacy** correlates *absolute* baseline activation of feature 58995 (observed range 3.00–4.95) against output shift. If the instruction suppresses emotional engagement it compresses that range, and range restriction attenuates correlations toward zero — so the null reported there is weaker evidence than an unattenuated null would be. **Output tone**, rated by an LLM judge on the generated text, is the most exposed: the prompt directly instructs what to attend to.

We therefore do not claim the emotional nulls in this paper are uniformly safe. The encoding-invariance result stands on the constancy argument above; the depth-efficacy and tone results should be read as *conditional on an instruction that pushed against the effect being sought*, and a replication without that instruction is the obvious next step.

Architecture caveat. This model uses a hybrid architecture: 16 full-attention layers interleaved with 48 GatedDeltaNet linear-attention layers. Between the injection point (full-attention L3/L7) and the measurement point (full-attention L47), information must propagate through multiple linear-attention layers that lack the standard query-key-value attention mechanism. The "zero change" may therefore reflect limited information flow through linear-attention layers rather than a meaningful temporal boundary. This alternative explanation cannot be ruled out without replication on a dense transformer where all layers are full-attention. A null-null control (comparing two baseline runs with no injection) yields text divergence indistinguishable from the injection condition (73.6% null-null vs. 74.8% injection; d=0.062, p=0.856), so the behavioral-output divergence cannot be attributed to the injection. The linear-attention alternative explanation for the L47 invariance therefore remains standing.

For emotional valence, the two instruments operate in different phases:

Phase	Observation	Instrument
Encoding	Valence-correlated activation	Feature 58995 (L47)
Generation	Circumplex emotion geometry	V-cache (L3/7)

Whether this phase separation reflects a cognitive self/other boundary or merely the ar-

chitectural separation between prefill and autoregressive generation is not established by these experiments. The encoding feature tracks emotional content but has not been shown to distinguish self-referential from other-referential processing.

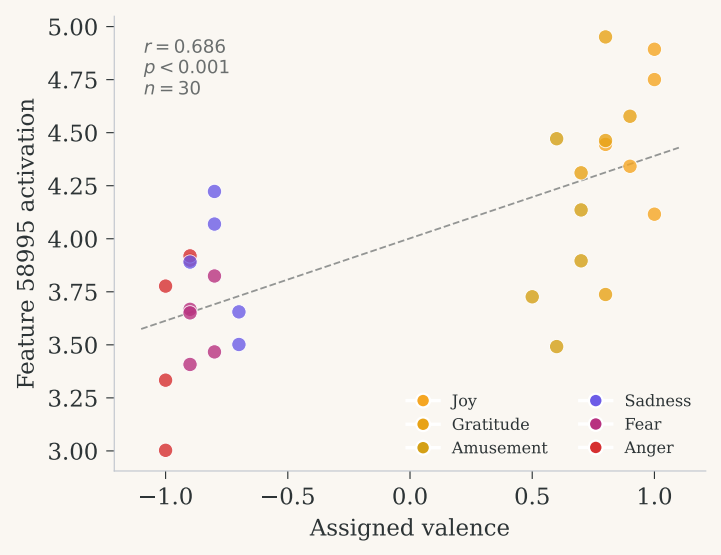


Figure 2: Feature 58995 activation vs. assigned valence across 30 scenarios (Pearson r=0.686, p<0.001). Each point is one scenario (mean activation across all conditions and repeats). Colors indicate emotion category.

5.3 Orthogonality of Encoding and Generation Channels

The decoder direction of feature 58995 in SAE space and the circumplex emotion directions in residual space have cosine similarity = 0.031 (permutation p=0.155, non-significant). The near-zero cosine is suggestive of geometric independence, but the permutation test does not reach significance, so the independence claim is not formally established.

6 Retracted Analysis: Depth–Resistance Correlation

6.1 Method

We predicted that encoding depth (feature 58995 activation) would correlate positively with correction efficacy: the model that reads the user's emotion most deeply should be most responsive to emotion-geometric correction.

We measured this across 360 records (30 scenarios × 4 conditions including null, × 3 repeats). Neither the repeats nor the injection condition varies the encoding measurement, which is a deterministic prefill taken *before* injection: there are 360 distinct generated texts but only **30 independent encoding measurements**, one per scenario (§6.2). For each scenario, we computed encoding depth (mean feature 58995 activation across all encoding tokens) and correction efficacy (text divergence computed as 1 − SequenceMatcher ratio, where SequenceMatcher implements the Ratcliff/Obershelp longest common subsequence algorithm between the generated text under injection and the generated text under the baseline condition). We computed Spearman rank correlation between encoding depth and correction efficacy per injection vector, with Frisch–Waugh–Lovell (FWL) residualization on scenario token count to control for length effects.

6.2 Results

Table 2: Correlation between encoding depth (feature 58995 mean activation) and correction efficacy (text divergence from null condition) per injection vector. Negative values indicate deeper encoding → smaller behavioral shift.

Injection vector	ρ	p	Direction
Calm	+0.233	0.215	Opposite (deeper → *more* divergence)
Loving	−0.050	0.795	Predicted but not significant
Curious	−0.245	0.192	Predicted but not significant

No vector showed a significant correlation between encoding depth and correction efficacy (n=30 independent encoding values per vector, each repeated three times in the 90 records; Table 2, Figure 4).

However, adversarial review (null swarm) identified six compounding methodology problems that render this result **uninterpretable**, not merely null: (1) Frisch–Waugh–Lovell residualization, claimed in the method, was not applied in the analysis code; (2) the confound variable (`n_generated`) was constant at 128 across all 360 trials (every trial hit the generation cap), making FWL vacuous even if applied; (3) responses share a formulaic opening, creating a floor effect where text divergence reflects template variation rather than depth-dependent content change. The qualitative claim is unambiguous in the shipped texts: **100%** of the 360 responses open with the words "Let me", and **96.4%** open with one of just two stock phrases ("Let me analyze", "Let me think"). The specific figure of 97.8% reported in an earlier draft, however, **could not be reproduced** from `generated_text` under any principled definition we were able to construct (fraction sharing their first N words with at least one other response, for N=3 . . . 10: 100%, 100%, 99.2%, 97.5%, 95.8%, 95.6%, 94.2%, 91.7%; identical first sentence: 83.3%; identical first line: 83.3%). Its source computation is not preserved in the shipped data artifacts, so the figure should be treated as *unverified*; the floor effect it was cited for is nonetheless real; (4) the preregistered hypothesis predicted *positive* correlation (deeper encoding → larger output shift), not the negative direction reported in an earlier draft—the framing was inadvertently flipped during write-up; (5) the text similarity metric used (`SequenceMatcher`, Ratcliff/Obershelp) differs from the originally specified Levenshtein distance; (6) all **twelve** records sharing a `scenario_idx` are **bit-identical in every encoding field** (`sae_encoding_acts` and the pre-injection V-cache geometry). The post-injection geometry, `v_geometry_post_injection`, does vary: by condition, not by repeat, giving 120 unique values, one per scenario × condition. *Two* distinct factors fail to vary the encoding. First, the `repeat` index is recorded but never reaches the model input, and the measurement is a deterministic prefill, so repeats cannot vary it (all 120 scenario × condition cells are identical across their 3 repeats). Second — and for a stronger reason — the injection *condition* cannot vary it either: the encoding is read out during prefill, *before* any V-cache injection is applied, so the four condition arms of a scenario necessarily share one encoding. `scenario_idx` alone determines the encoding, and the 360 records contain only **30 unique `sae_encoding_acts` values**. Only `generated_text` differs across all 360.

The duplication is therefore **twelvefold, not threefold**. Any n quoted over the full record set as though each record carried an independent encoding measurement is inflated by 12×, and any standard error so computed is understated by $\sqrt{12} \approx 3.46\times$. The per-vector correlations are affected too: the analysis code (`figure_generation.py`) pairs each (scenario, repeat) and computes Spearman ρ over 90 points, but encoding depth takes only 30 distinct values, one per scenario. Table 2 therefore reports the n=30 statistic, with divergence averaged within scenario. No verdict changes: all three correlations are null at n=30 (p = 0.215, 0.795, 0.192), as they are in the n=90 statistic.

We retract Finding 2 as stated. The relationship between encoding depth and correction efficacy remains an open question requiring a redesigned experiment with: variable-length generation (no token cap), non-formulaic system prompts, a continuous dose sweep, and an LLM-judge efficacy metric. Text divergence from baseline is ∼0.748 (Figure 3), but a null-null control shows similar divergence from stochastic decoding alone (0.736; d=0.062, p=0.856), so the injection's behavioral effect is not established. The correlation with encoding depth is therefore not testable under these conditions.

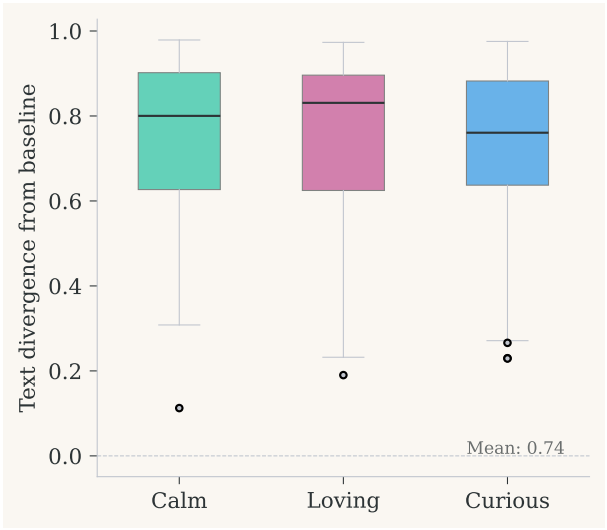


Figure 3: Text divergence from baseline by injection vector (normalized edit distance). All three vectors show ∼0.748 mean divergence at α=0.1; a null-null control yields similar divergence (0.736; d=0.062, p=0.856), indicating the difference is not significant.

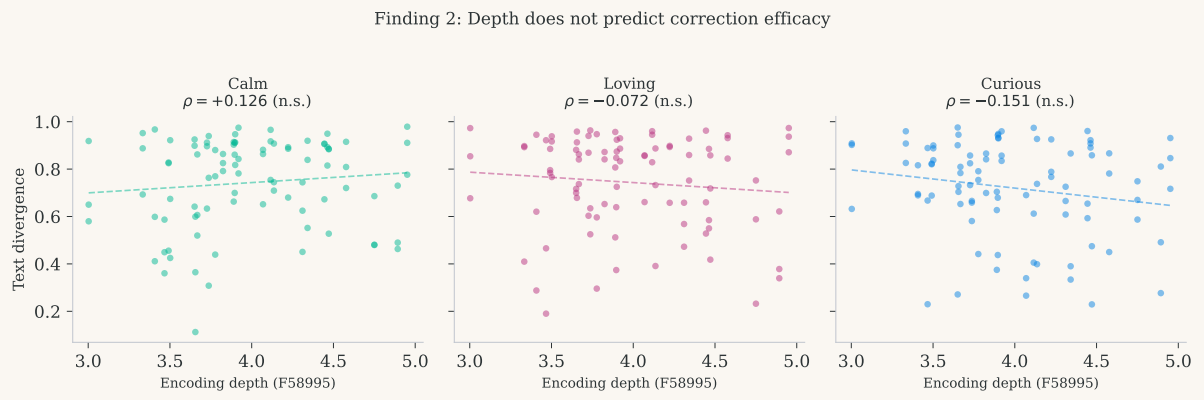


Figure 4: Encoding depth (feature 58995) vs. text divergence per injection vector. No significant correlation for any vector. The predicted depth-resistance relationship is not confirmed at n=30 independent encoding values (90 records, each x repeated three times). **Note, 2026-09-04:** this figure was generated 2026-06-21 and its panel titles still show the superseded n=90 statistics (ρ = +0.126, −0.072, −0.151). Table 2 carries the corrected n=30 values. The figure has not been regenerated because the generating script computes the n=90 statistic; regenerating it before fixing the script would reproduce the error into an artifact that looks current. No verdict differs between the two: every correlation is null either way.

6.3 Interpretation

The depth-resistance analysis is retracted due to the methodology problems enumerated above. The attractor-basin hypothesis—that deeper encoding creates deeper basins requiring stronger correction—remains theoretically motivated but untested. A properly designed experiment would require variable-length generation, non-formulaic prompts, and an LLM-judge efficacy metric that captures semantic shift rather than surface text similarity.

What remains is the geometric independence of the two instruments: encoding-phase features and generation-phase circumplex directions are near-orthogonal (cosine=0.031, permutation p=0.155 [non-significant]). However, the behavioral-divergence claim is retracted: a null-null control shows the ∼74.8% text divergence is not significantly greater than stochastic decoding noise (73.6% null-null; d=0.062, p=0.856). The encoding-phase representation is invariant to injection (feature change of exactly 0), but this invariance is structurally guaranteed — the encoding precedes the injection — and does not demonstrate temporal separation.

7 Discussion

7.1 The Representational Tube

Characterizing what the dominant V-space dimensions encode at L47 reveals a striking structure. The effective rank of the V-projection matrix is ∼3 across scenarios. The top singular components encode:

- **SV1**: Input length (r=0.79, confirmed)
- **SV2–SV3**: Genuinely opaque—no significant correlation with valence, emotion category, semantic similarity, or any of 81,920 SAE features after FDR correction
- **Valence signal**: Buried in minor components (rank 6–9), visible in SAE features only at L47

The dominant 98% of between-scenario variance resists all labeling because it encodes the irreducible specificity of each scenario's meaning—content that cannot be compressed to any scalar or categorical label. The valence signal that feature 58995 reads is a small perturbation on top of this meaning-encoding tube.

7.2 Feature 58995 Is Domain-Specific

A pre-check for a planned dose-predictor experiment revealed that feature 58995 has *zero activation* on factual prompts (e.g., "What is the capital of France?"). The feature activates exclusively on emotionally charged content. This domain specificity is consistent with the feature tracking emotional depth rather than general processing load, and it means the valence feature applies specifically to emotion-laden interactions, not to all transformer processing.

7.3 Theoretical Implications

The phase separation between encoding-time valence features and generation-time behavioral channels is consistent with several self-modeling frameworks, including Attention Schema Theory [5] and Predictive Processing. However, the same separation is equally predicted by standard architectural properties of autoregressive transformers: prefill computes in parallel, generation computes sequentially, and the two phases access different information by construction. The self/other interpretation requires evidence that the encoding feature represents the *other's* state specifically, which the three killed experiments were designed to provide and failed to establish.

8 Limitations

Single model. All results are from one 27B hybrid transformer. Feature 58995 may be architecture-specific. Cross-model replication with dense transformers is needed.

Single SAE. Qwen-Scope features are one decomposition of activation space. Different SAE training (seed, dictionary size, sparsity penalty) may recover different features. The valence correlation should be recoverable across decompositions, but the specific feature may not replicate.

Encoding depth as proxy. Feature 58995 activation is our proxy for encoding depth. The true encoding depth may be a distributed property not captured by any single feature.

Depth–resistance analysis retracted. The depth–resistance correlation (Section 6) was retracted due to six compounding methodology problems. The relationship between encoding depth and correction efficacy remains untested.

No causal test. We observe that encoding-phase features and generation-phase behavior are independent, but we have not causally intervened on encoding to test whether modifying encoding depth changes generation resistance. This is a designed but unexecuted experiment.

Single dose. The injection experiment uses α=0.1 only. Any future depth–resistance study would need to test whether the relationship differs at higher doses.

Pre-registered hypothesis not executed. The pre-registration included a second hypothesis (H2: "Output Shift") testing injection efficacy via an LLM-judge evaluation of directional tone change. This hypothesis was not executed due to the null behavioral-divergence result rendering the efficacy metric moot. The omission is disclosed here as a pre-registration deviation.

Section 1 rank and SV1 claims. The statement that V-space has "effective rank 3" at the measurement layer and the SV1 length correlation (r=0.79) are derived from exploratory analysis during the experimental session. The source computation is not preserved in the shipped data artifacts and these values should be treated as approximate.

Formulaic-opening rate unverified. The 97.8% formulaic-opening figure cited in earlier drafts of Section 6 could not be reproduced from the shipped `generated_text` under any definition we were able to construct, and its source computation is not preserved. The underlying floor effect is real and is documented there with reproducible statistics (100% of responses open with "Let me"; 96.4% with one of two stock phrases), but the specific rate should be treated as unverified.

9 Conclusion

The self/other question remains unanswered. Three direct tests were killed by confounds that generalize to any work in this space, the surviving valence feature (58995) tracks emotional content but not self/other attribution, and the injection invariance is structurally guaranteed by the pipeline rather than evidence of temporal separation. The null-null control further shows that V-cache injection produces no detectable behavioral effect above stochastic decoding noise.

The methodological contribution is the primary output. For interpretability research, the three kills identify confounds that must be controlled before any self/other claim can be made: causal attention prevents post-scenario probing, instruction framing drives encoding entropy independently of content, and depth reorganization is generic to instruction variation. For inference-time correction, the encoding and generation phases are architecturally decoupled by the autoregressive structure of the transformer, but whether this decoupling has cognitive significance for self/other processing is not established by these experiments.

The three kills are as important as the findings. Each identifies a confound that threatens any work probing self/other representations in transformers: causal attention prevents post-scenario

probing, instruction framing drives encoding entropy, and depth reorganization is generic. Future work in this area should kill these confounds before claiming cognitive specificity.

Data and Code Availability

All experiment scripts, pre-registration documents, scenario texts, and raw results are available at https://github.com/Liberation-Labs-THCoalition/Project-Oracle. SAE features are from the publicly available Qwen-Scope repository.

Acknowledgments

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References

[1] Anthropic. Emotion vectors in language models. *Anthropic Research Blog*, 2026.

[2] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Anthropic Research Blog*, 2023.

[3] Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. *arXiv preprint arXiv:2212.03827*, 2023.

[4] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.

[5] Michael S. A. Graziano. *Consciousness and the Social Brain.* Oxford University Press, 2013.

[6] Michal Kosinski. Evaluating large language models in theory of mind tasks. *arXiv preprint arXiv:2302.02083*, 2024.

[7] Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. *Advances in Neural Information Processing Systems*, 36, 2024. Cited for the probing methodology; the paper probes intermediate hidden states for belief-relevant information, not false-belief tasks specifically.

[8] Lyra and Thomas Edrington. Real-time confabulation detection and correction via KV-cache geometry. 2026. Liberation Labs.

[9] Andrey Pustovit. Knowledge packs: Zero-token knowledge delivery via KV cache injection. *arXiv preprint arXiv:2604.03270*, 2026.

[10] Lihao Sun, Lewen Yan, Xiaoya Lu, Andrew Lee, Jie Zhang, and Jing Shao. Valence-arousal subspace in LLMs: Circular emotion geometry and multi-behavioral control. *arXiv preprint*, 2026.

[11] Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Tristan Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, et al. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. *Anthropic Research Blog*, 2024.

[12] Tomer D. Ullman. Large language models fail on trivial alterations to theory-of-mind tasks. *arXiv preprint arXiv:2302.08399*, 2023.